

Chapter 36 — Pediatric Urogenital Imaging

- **Age-specific differences:** studies routine and tolerable in adults may need **anaesthesia/sedation** or specially trained technicians in children; doses must be adjusted and equipment sized for the smaller patient — the urologist must know these differences to order and interpret tests.
- **Weighing the decision to image:** radiation dose per study, likelihood of **interval follow-up** (further exposure), need for contrast and need for sedation → set against the anticipated clinical benefit.
- **ALARA ("as low as reasonably achievable") principle** is the consensus standard (International Commission on Radiological Protection, ICRP). Individual radiosensitivity cannot be predicted, so **every patient is treated as radiation sensitive** — especially in pregnancy, fetal life and childhood.
- **Guidelines on imaging selection** come from the American Academy of Pediatrics (AAP), National Institute for Health and Care Excellence (NICE), Italian Society of Pediatric Nephrology, Canadian Paediatric Society (CPS), Polish Society of Pediatric Nephrology (PSPN) and European Association of Urology/European Society for Paediatric Urology (EAU/ESPU) — all driven by the aim of **reducing unnecessary invasive testing**, built on best available data plus expert opinion.
- **American College of Radiology (ACR) Appropriateness Criteria** — Haematuria–Child (2018) and Urinary Tract Infection–Child (2024): balance **radiation exposure against diagnostic yield**, based on the prevalence of findings in specific clinical scenarios.
- **Organising frame of the chapter:** non-ionising then ionising modalities; for each, three levels (kidney/ureter; bladder/urethra; scrotum/testes/internal genitalia) and five indications (**obstruction → cysts/masses → infection → other congenital malformations → trauma**).

TECHNIQUE

Non-ionising

- **Sonography is the archetypal minimally invasive study** — hence its widespread use.
- **Magnetic resonance (MR) is not minimally invasive in practice:** long acquisition times mean many children need **sedation or anaesthesia**.
- **Sedation/anaesthesia risk usually exceeds contrast risk:** it is typically higher than the risk of **nephrogenic systemic fibrosis** from gadolinium (poor renal function, or immature neonatal renal function) and of **iodinated-contrast nephropathy** in poor renal function.
- **Allergic reaction:** both contrast classes carry a higher (but still very small) risk.
- **Radiofrequency heating:** MR radiofrequency pulses can cause patient heating → must be monitored; **hyperthermia** reported rarely.
- **Long-term MR safety** questioned because magnetic-field exposure alters gene expression experimentally; no human data, but **fetal MR is avoided in the first trimester**.
- **Ultrasonography** has shown no gene-expression changes → regarded as **safe at all ages and all stages of gestation**.

Ionising Radiation

- **More harmful in children than adults** because of (1) greater **radiosensitivity of developing tissues** and (2) potential for higher **cumulative lifetime dose**.
- **Stochastic effects** of repeated exposure (internal cell injury) → **radiation-induced malignancy after a long latent period**.

- **Reading the risk literature:** determine whether risk is **estimated, modelled or measured**; measured studies are confounded by technique, age, size and intervening studies over long intervals.
- Despite these limits, there is reasonable evidence of a **dose-related risk in children and even in utero**.
- **Responses:** dose reduction per test while preserving diagnostic accuracy; better decision making; **non-ionising alternatives**.

Concept — *Every paediatric imaging choice is a trade between three harms — **ionising radiation, sedation and contrast** — against diagnostic yield; the child's long remaining lifespan is what tips the balance toward sonography and ALARA.*

ULTRASONOGRAPHY

Fundamentals of Sonography in the Fetus and Child

- **Prenatal sonography** can diagnose many urological conditions with near certainty by assessing: presence and quality of renal cortex → laterality → kidney position and relation to other organs → **amniotic fluid volume** and urine in the bladder.
- **Amniotic fluid origin:** before **8–10 weeks** it is a dialysate of fetal blood across permeable fetal skin; fetal urine production begins at **10–12 weeks**; by **20 weeks** sonography detects most urological abnormalities.
- **Normal urinary tract:** two kidneys in the renal fossae; cortex **isoechoic or slightly hypoechoic to liver**; discrete corticomedullary interfaces; no cortical cysts, masses, or dilatation of collecting system or bladder.
- **Fetal bladder** should be visible in all fetuses by **10–12 weeks**; cycling may be seen with patience; it **never empties completely** (small residual always present).
- **Umbilical cord placement and anterior abdominal wall integrity** must be checked — abnormalities are highly associated with **bladder exstrophy and cloacal exstrophy**.
- **Newborn kidney trap:** distinct corticomedullary differentiation (hypoechoic pyramids) plus slightly hyperechoic parenchyma can be **mistaken for hydronephrosis**.
- **Hydronephrosis is underestimated** on an early postnatal scan (**up to 2 days after birth**) and in dehydration.
- **Prenatal sonography limitations:** low amniotic fluid, maternal habitus, maternal bowel gas, maternal pelvic bones, fetal position — in these respects **MRI is superior**.

Figure 36.1 — Newborn versus older-child kidney. High-contrast corticomedullary differentiation of the neonatal kidney mimics dilated calyces; compare with the less differentiated older-child kidney.

Kidney/Ureter

Obstruction

Hydronephrosis versus cystic renal diseases.

- **Commonest prenatal sonographic findings:** collecting-system dilatation and parenchymal cysts.
- **Differential:** hydronephrosis (obstructive and non-obstructive), **multicystic dysplastic kidney (MCDK)**, polycystic kidney disease (autosomal recessive and autosomal dominant), cystic nephroma.
- **Reporting is not standardised** across specialties. The **Society for Fetal Urology (SFU) classification** is most used by paediatric urologists — based on calyceal dilatation, minor-calyx involvement and parenchymal thinning. Others use renal pelvic **anteroposterior diameter** or combined systems.
- **Because reporting varies, the clinician must review the actual images** before deciding.

- **Natural history:** most hydronephrosis is **benign and improves or resolves**. Profound developmental abnormalities present as moderate–severe hydronephrosis from high-grade obstruction; the **transition from dilated to non-dilated tract localises the obstruction** — ureteropelvic junction (UPJ), ureterovesical junction (UVJ) or urethra.

FIGURE 36.2 — SOCIETY FOR FETAL UROLOGY GRADING (POSTNATAL SONOGRAPHY)

SFU grade	Sonographic criterion
0	No central renal dilatation
1	Renal pelvis only visible
2	Major calyces identified
3	Major and minor calyces identified
4	Grade 3 features plus parenchymal thinning

- **Grade 3 is heterogeneous:** many degrees of collecting-system dilatation satisfy its criteria — a reason to look at the images, not the grade.
- **Grade 3 → 4 step** is defined by the parenchyma, not by further dilatation.

Dilated ureters.

- Usually a **hallmark of obstruction**, but ureteric dilatation also occurs **without obstruction**, e.g. vesicoureteral reflux (VUR).
- **Megaureter and ureterocele** have obstructive and non-obstructive variants → distinguish with a **functional drainage test** (diuresis renography or MR urography).

Functional interpretation of sonography of the kidney.

- Standard ultrasound is **anatomic**; function is only inferred.
- **Key clinical question in hydronephrosis:** obstruction vs VUR vs normal variant (neither).
- **Bilateral hydroureteronephrosis + thickened bladder + poor emptying → voiding cystourethrogram (VCUG)** to exclude **posterior urethral valve (PUV)**.
- Otherwise, **diuretic renography is the gold standard** for evaluating hydronephrosis.
- **Emerging:** computer-aided shape analysis and **ultrasound elastography** (parenchymal/pelvic tension as indirect measure of obstruction) — if validated, could greatly reduce ionising functional studies.

Figure 36.3 — Posterior urethral valve on sonography.

- **Renal signs:** echogenic parenchyma, moderate–severe hydronephrosis, **renal cortical cyst** (renal dysplasia); **bladder view:** marked bilateral distal hydroureter.

Cysts/Masses/Stones

Renal cysts. Cysts can mimic hydronephrosis but are distinguished because **cysts do not communicate**, whereas dilated calyces communicate with the renal pelvis.

COMPARISON — CYSTIC RENAL LESIONS ON SONOGRAPHY

Lesion	Laterality	Sonographic hallmark	Age / note
MCDK	Unilateral	Non-communicating cysts of varying size, randomly distributed ; scant hyperechoic parenchyma	Prenatal/ neonatal
Autosomal recessive polycystic kidney disease (ARPKD)	Bilateral	Enlarged, homogeneously hyperechoic parenchyma (dilated collecting tubules)	Prenatal

Lesion	Laterality	Sonographic hallmark	Age / note
Autosomal dominant polycystic kidney disease (ADPKD)	Bilateral	Enlarged kidneys, parenchyma almost replaced by cysts	—
Multilocular cystic nephroma	Usually unilateral	Non-communicating cysts of varying size, more parenchyma than MCDK	Benign; tumour of infancy
Cystic Wilms tumour	Usually unilateral	Shares features with MCDK and cystic nephroma	2–4 years

- **Laterality is the first discriminator:** unilateral random cysts → MCDK; bilateral → think polycystic disease.
- **Trap:** ARPKD appears as **large, bright kidneys rather than discrete cysts** — the echogenicity comes from dilated collecting tubules.
- **Age separates cystic nephroma (infancy) from cystic Wilms (toddler).**

Renal and abdominal solid masses.

- **Differential:** neuroblastoma, congenital mesoblastic nephroma, Wilms tumour, renal vein thrombosis, renal artery thrombosis, adrenal haemorrhage, rare solid tumours (rhabdoid, clear cell, angiomyolipoma).
- **Sonographic hallmarks + age + presentation** usually suggest the diagnosis, but **contrast-enhanced computed tomography (CT)** commonly follows.

FIGURE 36.5 — RENAL "ENLARGEMENT": DIAGNOSIS SUGGESTED BY CLINICAL PRESENTATION

Age	Presentation	Diagnosis
1 month	Prenatally detected renal mass	Congenital mesoblastic nephroma (confirmed at nephrectomy)
17 months	Lower-pole mass on examination	Wilms tumour
4 days	Acute kidney injury + hypertension , diffuse renal enlargement	Renal vein thrombosis
1 week	Prenatal suprarenal mass	Adrenal haemorrhage

- **Pattern:** neonatal solid renal mass → mesoblastic nephroma; toddler → Wilms; neonate with diffuse enlargement, AKI and hypertension → vascular (renal vein thrombosis) rather than tumour.

Renal stones.

- Stone-forming children face a **lifetime of imaging** → minimise radiation; inform parent and older child so they can advocate for this plan at unfamiliar emergency facilities.
- **Sonography may suffice** for initial screening and follow-up in most children with acute renal colic, rather than reflex CT or kidney-ureter-bladder (KUB) radiograph.
- Ideal for **hydronephrosis/hydroureter**, directs further imaging if the stone is not seen; reliably detects **intraparenchymal calculi and nephrocalcinosis**.
- **Calculus = echogenic with acoustic shadowing** vs **parapelvic fat = echogenic without shadowing**.
- **"Twinkling" artefact** on colour Doppler distinguishes calculi from other hyperechoic foci.

Ureteral stones.

- Proximal ureter may or may not be dilated, depending on the **degree of obstruction**.
- **Distal/pelvic ureteric stones are easier to image in children than adults** (shorter distance to the stone); twinkling also seen.

Infection

- **Role:** initial screening for obvious anatomical abnormality; not routine at onset of treatment, but **failure to improve or worsening on antibiotics → sonography**.
- **Sensitive for significant renal abnormality**, but much less for uncomplicated VUR, duplication anomalies, **acute pyelonephritis (APN)** and focal scarring. Alone, rarely useful in most febrile and non-febrile urinary tract infection (UTI) → often **combined with VCUG** when a urological abnormality is suspected.
- **Imaging algorithms** are mostly applied after treatment, to identify risk factors for recurrence and anomalies predisposing to complicated UTI and renal injury.
- **Hospitalised child:** screening sonography **before discharge to exclude obstruction**; otherwise imaging is deferred until after discharge.
- **Hydronephrosis without VUR → diuresis renography**; the combination reliably localises the obstruction.
- **Grey-scale sonography is inferior to technetium-99m dimercaptosuccinic acid ($^{99m}\text{Tc-DMSA}$)** for APN and scarring.
- **A normal sonogram neither risk-stratifies a child with APN nor predicts VUR.**

Contrast-enhanced urosonography (CEUS).

- **Agent:** gas-containing **sulfur hexafluoride lipid-type A microspheres** (Lumason/SonoVue); US Food and Drug Administration (FDA)-approved for **intravenous or intravesical** use.
- **Contrast-enhanced voiding urosonography (CeVUS)** — commonest application: VUR detection, with a **five-point grading system analogous to VCUG**.
- **Cost:** initially higher (time-consuming), offset by being a **non-ionising method for detecting and monitoring VUR**.
- **Single-centre retrospective paired comparison with VCUG:** moderate sensitivity and specificity for VUR, with **identical performance between two readers** (good inter-rater reliability).
- **Systematic review/meta-analysis** (Table 36.1): first- and second-generation agents equally accurate, **overall false-negative rate 3%** → authors recommend CeVUS for VUR detection **without reservation**.
- CeVUS may also replace contrast cystography for VUR **after renal transplantation** and during **video-urodynamics**.

TABLE 36.1 ACCURACY OF CEVUS FOR THE DETECTION OF VUR: A SYSTEMATIC REVIEW OF 2768 CHILDREN FROM 36 STUDIES

Contrast agent	AUC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
First generation	0.97 (0.95–0.98)	0.92 (0.86–0.96)	0.94
Second generation	0.97 (0.95–0.98)	0.93 (0.86–0.97)	0.91 (0.86–0.95)

Yousefifard M, Toloui A, Rafiei Alavi SN, et al. J Pediatr Urol 2022;18(1):61–74. AUC, area under the receiver-operating-characteristic curve; CI, confidence interval.

- **What the table is for:** VCUG is the reference standard; CeVUS matches it (AUC 0.97) whichever agent generation is used.
- **Pattern:** the newer agent does not buy accuracy — both generations perform the same.
- **Trap:** "non-ionising = less accurate" — here it does not; the limitation is cost/time and grading comparability, not detection.

Intrarenal reflux (IRR) on CeVUS.

- **IRR marks renal regions at risk of injury from APN**; applying CeVUS to the **"top-down"** approach is attractive.

- **Bottom lines from recent series:** CeVUS detects IRR in a minority of renal units, with **excellent concordance with VCUG**; dilating VUR and IRR are both **commoner in males**; IRR is seen mostly **during voiding**, most often in the **lower pole**, **rises with each VUR grade** and is **associated with renal scarring**.
- Polar distribution of IRR on CeVUS **mirrors the distribution of acute and chronic DMSA lesions** — IRR is the anatomical substrate of pyelonephritic scarring.

Trauma

- Sonography compares favourably with **contrast-enhanced CT** for the initial evaluation of renal injury in acute paediatric trauma.
- **Contrast-enhanced sonography is as sensitive as contrast-enhanced CT** for paediatric blunt abdominal trauma.

Bladder and Urethra

Obstruction

- **Anatomical detection:** ureteroceles, large diverticula, bladder calculi, debris, suspicious masses (Figure 36.7: ureterocele, diverticulum, calculi with twinkling).
- **Bladder wall thickness** varies with volume and age; volume-indexed standards are too complex/variable for wide use — but a **grossly thickened bladder with a voiding-dysfunction picture is suspicious**.
- **Basic function:** emptying quantified by **prevoid and postvoid volumes**.
- **"Keyhole sign"** = thickened bladder + dilated posterior urethra → **highly suggestive of PUV** (Figure 36.8: prenatal bilateral hydroureteronephrosis + thick bladder + keyhole = bladder outlet obstruction).

Cysts/Masses

Midline pelvic cysts.

- **Differential:** hydrometrocolpos (urogenital sinus anomalies), ovarian cyst, distended bladder, urinary ascites, **dilated prostatic utricle**.
- **Hydrometrocolpos** = uterus and vagina distended with mucus or blood, from vaginal obstruction (**imperforate hymen, vaginal atresia, transverse vaginal septum**) or **retrograde urine flow** in urogenital sinus and cloacal malformations.
- **Dilated prostatic utricle** = remnant of the müllerian duct opening → fluid-filled midline structure **posterior to the bladder in males**.

Figure 36.9 — Cloacal anomaly.

- **Oligohydramnios + dual pelvic cysts below a dilated bladder** (34-week female fetus) → postnatal sonography shows **hemivaginas** → contrast via the **single perineal opening** opacifies both hemivaginas, bladder and distal colon = cloaca.

Bladder tumours.

- **Bladder as full as possible** to assess contents and wall.
- **Masses:** clot, benign and malignant tumours, foreign bodies, pseudotumours.
- **Clot vs tumour** (child with gross haematuria): image **supine then prone** → **clot moves, tumour stays fixed**.
- Wall tumours arise from urothelium, mesenchyme, neuroendocrine cells or in response to viral infection; **sonography does not discriminate histology** → further clinical assessment.

Infection

- **Viral cystitis:** **thickened bladder wall**, sometimes enough to mimic a mass; clinical triad → **viral upper respiratory prodrome + gross total haematuria + marked urgency/frequency**.

Other Congenital Malformations

Abdominal wall defects. Bladder exstrophy and prune belly syndrome — abdominal wall formation defects with urogenital anomalies.

COMPARISON — PRENATAL SONOGRAPHIC CLUES: CLASSIC BLADDER EXSTROPHY VS PRUNE BELLY SYNDROME VS POSTERIOR URETHRAL VALVE (WITH FIGURE 36.10)

Feature	Classic bladder exstrophy	Prune belly syndrome	Posterior urethral valve
Bladder	Not visualised	Distended, wall not thickened	Thickened wall + keyhole sign
Upper tracts	Normal kidneys, no hydroureter	Hydronephrosis/hydroureter	Hydronephrosis/hydroureter
Abdominal wall / cord	Irregular lower wall; low cord insertion	—	—
Genitalia	Short phallus	Testes absent from scrotum	—
Amniotic fluid	—	May be high (raised amniotic fluid index)	—

- **Trap:** prune belly and PUV share hydroureteronephrosis — the **thick bladder/keyhole** points to PUV, the **empty scrotum** to prune belly.

Trauma

- No bladder/urethral sonographic trauma content is given under this heading; lower-tract trauma is imaged with contrast studies (see VCUG and CT sections).

Scrotum/Testes/Internal Genitalia

Acute Scrotum

- **Genital imaging covers four conditions:** acute/enlarged scrotum, testicular/paratesticular masses, cryptorchidism, ambiguous genitalia. **Ultrasonography is always the initial and usually the only modality.**
- **Overuse is widespread** — most scrotal problems are diagnosed by history and examination; **avoid over-interpreting imaging that contradicts the clinical examination.**
- **Colour Doppler** helps when the examination is **equivocal for torsion**; **never delay exploration** for imaging if torsion is suspected and delay threatens salvage.
- **"Whirlpool sign"** — high-resolution imaging of the **twisted spermatic cord itself**, developed because colour Doppler gives false negatives; raises sensitivity to about **96% (from about 76% with colour Doppler)**, specificity about 99%.
- **Salvageability** is hard to predict except: **no arterial flow + parenchymal heterogeneity → all testes non-viable** at exploration.
- **Epididymo-orchitis:** hyperaemia of epididymis and/or testis — but must be correlated clinically because **recent spontaneous detorsion also gives hyperaemia.**
- **Torsion of testicular/epididymal appendage:** upper-pole/epididymal **hypoechoic avascular nodule with surrounding hyperaemia** (skilled operator).
- **Acute idiopathic scrotal oedema: "fountain sign"** — hyperaemia of a thickened scrotal wall with none of the above diagnoses.

Figure 36.11 — *Torsion.* Whirlpool sign in the right cord with **absent Doppler flow** vs ample flow on the non-torsed left side (19-year-old; incidental left varicocele).

Cysts/Masses

- Interpret with **age and full clinical picture**; generally well characterised by ultrasound.

- **Commonest prepubertal primary testicular tumour = benign teratoma** — heterogeneous: solid + cystic + calcified components.
- **Yolk sac tumour** — homogeneous, well-vascularised mass (Figure 36.13: enlarged testis in a 2-year-old).
- **Epidermoid cyst** — alternating hyper/hypoechoic "**onion rings**" with **no internal flow**.
- **Congenital adrenal hyperplasia: bilateral hypoechoic, hyperaemic, heterogeneous testicular adrenal rests** with inadequate steroid replacement or poor compliance.

Undescended Testis

- **Do not image routine cryptorchidism** (American Urological Association "Choosing Wisely") — **poor sensitivity** for detecting/localising the testis and it **does not change management**.
- **Reasonable exception: a difficult-to-examine obese child**, where finding an inguinal testis simplifies the surgical approach.
- **Primary modality in ambiguous genitalia or severe hypospadias with non-palpable gonads** → detects müllerian structures to guide work-up.

Differences of Sex Development

- Sonography is the basis for identifying **internal genital structures** in differences of sex development (DSD) — hypospadias with unilateral/bilateral non-palpable testes, or female genital anomalies.

Concept — Sonography is first-line at every level because it is **safe, repeatable and anatomic**; its weakness is **function** (obstruction, reflux, pyelonephritis), and the rest of the chapter is essentially about which test fills that functional gap at the least cost in radiation and sedation.

MAGNETIC RESONANCE IMAGING

Fundamentals of MRI in the Fetus and Child

- **Used to characterise complex congenital anomalies** when sonography lacks detail; **not helpful for infection, stones or trauma**.
- **Precontrast series may suffice** for ectopic ureters, duplication anomalies, renal cysts, calyceal diverticula.
- **Limits:** cost, access, **sedation** in young children.
- **Fetal MRI** is performed alongside a **third-level obstetric ultrasound after about 20 weeks**; any anatomy visible on ultrasound is more easily appreciated on MRI.

Protocol

- **MR urography (MRU) gives anatomy and function**. Sequence: **intravenous normal-saline hydration + Foley catheter → furosemide (before or during) → T2-weighted fast spin echo (FSE) sequences** (rapid single-shot for anatomy; longer 2D/3D FSE for resolution; 3D reconstructable) → **contrast → dynamic T1-weighted gradient-recalled echo** (vasculature, parenchymal enhancement, excretion) → **delayed images** for ureteric course.
- Hydration + diuretic → faster, better distension and **fewer artefacts** on contrast imaging.
- **Patlak method** — graphic analysis using **linear regression of tracer clearance** — is the usual functional analysis of dynamic data.

Sedation in the child.

- Standard protocol takes **up to 1 hour**; a shortened **~30-minute** format exists but needs larger validation.
- **Propofol conscious sedation or general anaesthesia** is often needed **under 6 years**.
- **Avoiding sedation:** respiratory-triggered (breath-hold) sequences — at the cost of longer studies; **child life specialists** (preparation and coping techniques).

Kidney/Ureter

Hydronephrosis/Obstruction

- **MRU is a plausible alternative to diuretic renography:** far better anatomical resolution, **no radiation**; pinpoints anomaly, gives **differential function and drainage** for surgical planning.
- Protocols resemble **mercaptoacetyltriglycine (MAG3)** diuretic renography; **gadolinium-based** contrast; well hydrated; **furosemide 1 mg/kg (max 20 mg)**; timing of furosemide, catheterisation and positioning vary by protocol.
- **Sequence:** precontrast T2 → **3D T2 fat-saturated, 1-mm slices** → slow contrast injection → T1 fat-saturated dynamic scan until contrast fills the ureters.
- **Barrier to adoption:** complex post-processing by a specialised clinician; free software (e.g. "chop-fmru", ImageJ from the National Institutes of Health) assists.
- **Status:** promising, but **equivalence to diuretic renography unproven** and interpretation training still needed.

Cysts/Masses/Stones

- **Poor for stones;** useful for **complex renal cysts and renal masses**.
- **MRI vs CT for paediatric renal tumours:** similar for **capsular penetration** and **lymph node metastasis**; **MRI better for synchronous contralateral lesions**.

Infection

- Gadolinium-enhanced MRI images APN with **accuracy similar to DMSA, single-photon emission CT (SPECT) and contrast CT**, but is **impractical** (long sedation) and rarely changes care.

Other Congenital Malformations

- **Complex anatomy** → MRU gives helpful anatomical detail.

Bladder/Urethra

- **Rarely more sensitive than a well-performed sonogram** for the lower tract, but **extremely useful for bladder masses, especially rhabdomyosarcoma**.
- **Sequences:** coronal wide-field short tau inversion recovery (STIR); diffusion-weighted imaging ($b = 0, 100$ or 600 s/mm^2); non-enhanced and gadolinium-enhanced pelvis; 3D volumetric T1; axial and sagittal T2 fat-saturated.
- **Urine masks T2-hyperintense lesions** → acquire non-enhanced T1 and fast sequences **before gadolinium** accumulates in tissue.
- **Rhabdomyosarcoma:** **uniformly low T1; heterogeneous high T2 and on contrast-enhanced T1 fat-saturated** images (heterogeneity attributed to necrosis).
- Useful for **staging**, but **no MRI feature predicts subtype**.

Scrotum/Testes/Internal Genitalia

Acute Scrotum

- **MRI has no role.**

Cysts/Masses

- **No role** in characterising scrotal cysts/masses; may be considered for **staging testicular cancer** — MR gives information predictably similar to the historical contrast CT of chest and abdomen.
- **Paediatric renal tumours:** MRI preferred for **locoregional staging**; CT preferred by some for **distant metastasis**.

Undescended Testis

- MR is **marginally better at best** than sonography for internal structures and gonadal detection.

Differences of Sex Development

- **Sonography + urogenitography usually suffice** for diagnosis and surgical planning; MRI adds anatomical information only in a **few very complex cases**.

Concept — MRI is the **non-ionising problem-solver** for complex anatomy, tumour staging and (potentially) functional obstruction studies; its ceiling is set not by image quality but by **sedation, cost and post-processing expertise**.

CONVENTIONAL RADIOGRAPHY

Plain Abdominal Radiography

- KUB used for **radiopaque stones** and for **constipation** in children with voiding dysfunction.
- **Few data support its accuracy** for diagnosing or monitoring constipation, but it gives **persuasive visual evidence** to sceptical parents.

Intravenous Pyelography

- **Intravenous pyelography (IVP) has fallen out of favour**, replaced by high-resolution anatomical and reproducible functional studies.
- **Technique**: sequential radiographs through excretion and drainage, in different focal planes.
- A good study localises **obstructive lesions and radiopaque stones**; drainage time assessed, but **less precisely than diuretic renography**.

Retrograde Pyelography

- Contrast injected into the ureteric orifice — **needs anaesthesia in a child**; best for **intraoperative localisation** of a lesion already found on other studies.
- **Trap**: free retrograde flow ≠ unobstructed antegrade flow (and vice versa), especially **after prior repair** — e.g. **small postsurgical tissue flaps** obstruct antegrade flow yet look unobstructed retrogradely.

Concept — Conventional radiography survives only where it answers a question nothing safer does; **retrograde flow is a valve-direction test**, so it cannot prove antegrade patency.

VOIDING CYSTOURETHROGRAPHY/MICTURATING CYSTOURETHROGRAM

Fundamentals of VCUG

- **Assesses lower urinary tract function**; routinely done **without sedation**, though invasive → pain/anxiety mitigation protocols exist.
- **After UTI**: defer until the child has had **several days of antibiotics and is clinically improving**.
- **Catheter check**: brief fluoroscopic image confirms tip in bladder and not too much catheter advanced.
- **Catheter not midline → malposition**: e.g. **utricle/posterior urethral abnormality in males, entry into an ectopic ureter in females**.
- Too much catheter → **knotting**.

Kidney/Ureter

Obstruction

- Routine VCUG for all hydronephrosis has been challenged; most algorithms reserve it for **hydronephrosis with dilated ureter(s)**.

Cysts/Masses

- **No renal cystic disease or mass** is specifically evaluated by VCUG.

Infection

- **Historical 4–6-week delay** after UTI (fear that trigonal inflammation causes transient VUR) is **unnecessary** — clinically significant VUR rarely disappears after treatment.
- **Early VCUG during febrile UTI: endotoxin-induced ureteric dilatation may overestimate the VUR grade.**

Other Congenital Malformations

- Only **UVJ anomalies** among upper-tract malformations benefit from VCUG.

Trauma

- **No renal or ureteric trauma** is specifically evaluated by VCUG.

Bladder/Urethra

Obstruction

- **Excellent high-resolution anatomical study:** bladder wall (trabeculation, ureterocele, diverticula, bladder-neck hypertrophy, tumours), urethra (**PUV**, stricture, diverticulum), bladder stones, rupture, foreign bodies; assesses emptying.
- **Scout film** can reveal many **spinal abnormalities**.

Figure 36.15 — VCUG findings.

- **Highly trabeculated bladder; ureterocele** seen within the bladder and then **everting**; large diverticulum with **elongated posterior urethra** in PUV; PUV with **bilateral grade 5 VUR**; **short distal bulbar stricture**.

Cysts/Masses

- Dilatations/masses seen: **urachal diverticula, bladder diverticula, dilated utricles, fibroepithelial polyps**.

Infection

- **Single or few non-febrile UTIs** (cystitis/urethritis) → VCUG adds little; with **increasing severity or recurrence**, some obtain VCUG to exclude anatomical risk factors.
- **Circumcised males:** UTI risk so low that **any UTI may merit evaluation**.

Other Congenital Malformations

Occult VUR.

- **Evidence for "occult" VUR:** new contralateral VUR appears in **7–33%** of unilateral VUR; VCUG finds VUR in only **about 40%** of DMSA-proven APN.
- **Mechanisms:** variation between voiding cycles; infection-induced UVJ dysfunction.
- **Positional instillation of contrast (PIC) cystography:** cystoscopically directed stream of contrast, **hung 1 m above the bladder**, aimed at the ureteric orifice to challenge it.
- **Indication:** **negative VCUG + recurrent febrile UTIs**; identifies occult VUR in up to **39%** with unilateral VUR and **59%** with negative VCUG + abnormal DMSA.
- **Debate:** critics — mostly **low-grade and transient**; proponents (with concomitant endoscopic treatment) cite a **58% reduction in recurrent UTI**.

Trauma

- **Pelvic trauma**, especially with **blood at the meatus and/or difficulty voiding** → contrast imaging of bladder and urethra; urethral injury most likely; **prompt intervention mandatory**.
- **Blood at meatus** → **retrograde urethrogram before any catheterisation** → urethra cleared → advance catheter → fill bladder by gravity until distended.
- **Expected bladder capacity (mL) = (age in years + 2) × 30, maximum 400 mL.**

Figure 36.16 — *Bladder injury with pelvic fracture*. Screening CT suggests injury → contrast cystography confirms: **extreme elevation of the bladder neck ("pie-in-the-sky")** and contrast collecting in the **retroperitoneal (extraperitoneal) space**.

Scrotum/Testes/Internal Genitalia

- **Not useful for routine genital anomalies**, except urethral, bladder and internal genital anatomy in **cloaca**, **urogenital sinus** and **DSD**.
- **Males with ectopic ureters or dilated utricles** may have recurrent UTI (**epididymitis**, febrile/symptomatic UTI) → **best evaluated with VCUG**.

Differences of Sex Development

- **Urogenitography** is essential for surgical planning and classification.
- **Technique**: catheter in the **single perineal opening** → contrast under fluoroscopy → define the **confluence of urethra and vagina** and orientation; alternatively Foley balloon inflated and pressed to the perineum for retrograde filling.
- Classified by **Shopfner's classification**.

Figure 36.18 — *Genitogram, 46,XX ovotesticular DSD*. **Low confluence** of urethra and vagina with reflux into the right fallopian tube → **Shopfner type III**.

Concept — VCUG is the **anatomical reference test for the bladder outlet and urethra**; its limits are that it is **invasive, ionising and a snapshot** — which is why reflux can be "occult" and why its timing after infection and its trigger criteria have been narrowed.

COMPUTED TOMOGRAPHY

Fundamentals of CT Imaging in Children

- CT's paediatric role rests on **availability and speed**, not superior diagnosis — best realised in **blunt abdominal trauma and polytrauma**; its need is **dwindling** as sonography and MR advance.

Kidney/Ureter

Obstruction

- **Little role in hydronephrosis** — little advantage over ultrasound, plus radiation.
- With IV contrast and delayed images, a **delayed nephrogram and ureteric drainage vs the contralateral side** signals obstruction, but is **hard to quantify**.
- Helpful anatomy in complex anomalies (ectopic kidney, ectopic ureter, megaureter, duplication, renovascular anomalies) when **sonography or MR are inconclusive or unavailable**.

Cysts/Masses

- **Contrast-enhanced CT** for paediatric renal tumours is **commonplace**.

Infection

- Limited role; reserved for when **non-ionising modalities are inconclusive**.

- CT and MR help in **renal abscess, extraurinary fistulisation, postoperative complications**, or infection secondary to obstruction.

*Figure 36.17 — Renal abscess. 6-year-old: abdominal pain, raised erythrocyte sedimentation rate and C-reactive protein, **inability to stand erect** → Doppler sonography shows heterogeneous lower-pole collection → **CT confirms abscess.***

Calculus Disease

- **CT's most obvious advantage** — better than sonography at detecting and quantifying stone burden, but the extra accuracy is **seldom clinically impactful**.
- Apply **ALARA**, especially in stone-formers facing lifelong repeated imaging.

Trauma

- CT with and without IV contrast is **routine in paediatric abdominal trauma**.
- **Timing trap**: images taken only in the **arterial phase** enhance cortex alone → show contusion/laceration but **miss collecting-system and ureteric injuries** → **delayed (excretory) images** are required.

Bladder/Urethra

- Little role in lower-tract obstruction, infection or malformation; may image bladder masses, but **MR preferred for soft-tissue resolution**.

Trauma

- Lower-tract CT reserved for **bladder or pelvic trauma**.
- **CT cystography**: antegrade/delayed filling or **retrograde catheter filling** → detects bladder or **augment rupture**.
- Advantage over conventional cystography: better anatomy and **immediate classification as intraperitoneal vs extraperitoneal**, which determines management.

Scrotum/Testes/Internal Genitalia

- **No role** in acute scrotum, genital cysts/masses, undescended testis or DSD; **part of staging for testicular/paratesticular malignancy** (metastases).

Concept — In children CT is chosen for **speed, not superiority**; outside trauma and tumour staging almost every CT question is answerable by sonography or MR without radiation — and when CT is used, **phase timing** decides whether the collecting system is actually seen.

DMSA RENAL CORTICAL SCINTIGRAPHY

- **Mechanism**: DMSA is taken up by **proximal tubular cells**, a process dependent on **renal blood flow**; APN reduces focal blood flow → **photon-deficient areas**.
- Lesions may be **focal, multifocal or diffuse**; distinguishing acute from chronic is difficult when **APN is superimposed on old scars**.
- **Validation (piglet autopsy model, histopathological reference)**: sensitivity **87%**, specificity **100%** for APN; SPECT gives slightly higher sensitivity, lower specificity, **equivalent accuracy** to planar (pinhole) imaging.

COMPARISON — ACUTE PYELONEPHRITIS VS RENAL SCAR ON DMSA

Feature	Acute pyelonephritis	Renal scar
Uptake	Decreased peripheral uptake (photopenia)	Decreased uptake
Renal contour	Reniform contour preserved	Volume loss → contour interrupted, concavity

Feature	Acute pyelonephritis	Renal scar
Timing	Seen reliably within days of the episode; resolves gradually over ~5 months	Any lesion still present at 6 months ≈ fixed scar

- **Timing follows the question:** scan **early (days)** to document APN; scan at **≥6 months** to document permanent scarring.
- **Acute lesions persist** (i.e. become scars) in **36–52%** of kidneys.

Figure 36.19 — *DMSA evolution*. Photopenic APN with preserved contour → either **scar formation** (contour lost) or **complete resolution by 6 months**.

Concept — *DMSA images tubular perfusion*, so inflammation and scar look alike on uptake; **contour and time** are what separate a reversible lesion from permanent damage.

MAG-3 DIURESIS-INDUCED RENAL SCINTIGRAPHY

- **Diuretic renography is the gold standard** for distinguishing obstructive from non-obstructive hydronephrosis/hydroureter; tracer ^{99m}Tc -diethylenetriamine pentaacetic acid (DTPA) or, more commonly, ^{99m}Tc -MAG3.
- **Strict protocol** required for reproducibility (the "well-tempered" diuretic renogram).

COMPARISON — THE FOUR KEY ELEMENTS OF A DIURETIC RENOGRAM

Element	How	Why / pitfall
1. Hydration	IV line for hydration + encourage oral fluids before arrival	Dehydrated patient responds poorly to diuresis → false obstruction
2. Region of interest (ROI) and background subtraction	Tight ROI round renal cortex (differential function) or round dilated collecting system (drainage); background ROI equally careful	Wrong portion of tract in ROI → spurious values
3. Bladder drainage	Urethral catheter	Removes effect of bladder filling on upper-tract drainage , confusion from VUR , and gonadal radiation from radioactive urine
4. Timing of diuretic	IV furosemide 1 mg/kg when the dilated system is maximally filled — not before	Early diuretic → drainage measured before system full

- **Outputs:** differential renal function, washout curves, washout half-times ($T_{1/2}$).
- **Predictors of progression** beyond $T_{1/2}$: **biphasic curve** (prompt drainage that stops) and **poor gravity-assisted (upright) drainage** → more likely to progress to high-grade obstruction.
- **Decide on:** age + renal function + diuresis $T_{1/2}$ + **curve shape** + **upright clearance** — and **review the actual images, ROIs and curves**.
- **No established paediatric $T_{1/2}$ cut-off** defines obstructed vs unobstructed → **repeat scans** during follow-up are often required.
- **Intermittent obstruction:** raised intrapelvic pressure lowers renal blood flow → **cortical retention of tracer** (akin to an obstructed nephrogram on IVP), visible **up to 48 hours** after the episode even when **Dietl's crisis** (nausea/vomiting + flank/abdominal pain) has resolved.
- **Versus sonography:** properly performed renography is **more reproducible and quantifiable**.

Figure 36.20 — *Hydration*. Child fasted overnight for sedation after bilateral pyeloplasty: despite fluid resuscitation, little urine → **poor cortical transit and poor drainage bilaterally**; repeated without fasting and with more aggressive hydration → brisk diuresis and **improved drainage half-times**.

Figure 36.21 — ROI and half-time calculation.

- If $T_{1/2}$ is timed from **tracer injection** (furosemide at 30 min), it wrongly includes cortical transit, secretion and accumulation time.
- **Correct $T_{1/2}$ = time for 50% of tracer to drain from the dilated collecting system** (from T_{max} /maximal filling).
- Changing ROIs showed the obstruction was at the **UVJ, not the UPJ** — ROI placement localises the level.

*Figure 36.22 — One renogram is not enough. Serial MAG3 studies in one patient: normal $T_{1/2}$ at 5 weeks and 9 months, then **markedly prolonged $T_{1/2}$ with heavy retention at 1 year** — progressive obstruction.**Figure 36.23 — Gravity-assisted drainage. Indeterminate bilateral drainage at 42 days; **15 minutes upright** cleared more on the left; at 8 months right function worsened with negligible diuretic and upright drainage, left drained spontaneously → **upright clearance discriminates the side that will progress**.**Figure 36.25 — Deferred decline. Prenatal hydronephrosis draining well at 3 months → **prolonged $T_{1/2}$ at 2 years** → pyeloplasty → afterwards **prompt spontaneous drainage, so furosemide not given**.**Figure 36.26 — Intermittent obstruction. Mild hydronephrosis with **rapid washout ($T_{1/2}$ 2 min)**, then flank pain and a **rising second limb** of the curve as tracer re-accumulates; after surgery function stable and drainage improved.**Figure 36.27 — Sonography may miss obstruction. SFU grade 3 (mostly intrarenal) dilatation, no hydroureter, preserved corticomedullary differentiation → MAG3 shows **high-grade UPJ obstruction with reduced function** and a continuously rising (non-draining) curve.*

Concept — The renogram measures **drainage under stated physiological conditions**, so every artefact (dehydration, wrong ROI, full bladder, mistimed diuretic) masquerades as obstruction; because no single $T_{1/2}$ threshold exists, the **trend over serial studies and the curve shape** carry the diagnosis.

RADIONUCLIDE CYSTOGRAPHY

COMPARISON — DIRECT VS INDIRECT RADIONUCLIDE CYSTOGRAPHY VS VCUG FOR VUR

Feature	Direct radionuclide cystography	Indirect radionuclide cystography	VCUG
Catheter	Required	Not required (after renography)	Required
VUR sensitivity	Greater than VCUG	Less than direct → not routine	Reference
Radiation	Lower than VCUG	Part of renography	Higher
Anatomical resolution	Poor (collecting system, urethra)	Poor	High
Physiology	Non-physiological (like VCUG)	Physiological , non-invasive	Non-physiological
Grading	Hard to match VCUG grades (mild/moderate/severe; Figure 36.28)	—	Five-grade system

- **Trade-off:** direct nuclear cystography buys sensitivity and lower dose at the cost of **anatomy**, and still needs a catheter.

RADIONUCLIDE TESTICULAR SCANNING

- In use since the **1970s**, to distinguish **torsion from inflammation** (epididymitis, orchitis, appendage torsion).
- **Technique:** IV ^{99m}Tc -pertechnetate → dynamic and static gamma images of the pelvis.
- **Photon-deficient = poor flow (torsion); photon-hyperdense = inflammation (epididymitis).**
- In experienced centres, sensitivity and specificity **up to 100%**, potentially better than sonography — but **limited availability, invasiveness and radiation** prevent adoption over ultrasound.
- **Best use:** cases **equivocal on examination and sonography**, and only if it does **not unreasonably delay** potentially testis-saving surgery.

KEY POINTS: IMAGING TECHNIQUES

- **ALARA** governs all paediatric imaging.
- **Non-ionising:** contrast-enhanced sonography and **machine-learning-enhanced** sonographic analysis should be adopted as they become available — may obviate more invasive tests.
- **Obstruction:** sonography is the **initial test and the follow-up test**; ancillary VCUG and/or diuresis renography per findings; normal infant kidney (hypoechoic pyramids, distinct corticomedullary junction) can be **mistaken for hydronephrosis**; diagnosing obstruction on renography requires review of **all test parameters**.
- **Cysts/masses/stones:** MCDK = randomly distributed cysts of varying size, scant hyperechoic parenchyma; ARPKD = bilateral, enlarged echogenic parenchyma; ADPKD = bilateral, parenchyma replaced by cysts; renal masses → **contrast-enhanced CT** follows sonography.
- **Infection:** guidelines use sonography as initial screen **despite low yield**; a normal sonogram does not risk-stratify febrile UTI nor predict VUR.
- **"Bottom-up" vs "top-down":** see comparison below.
- **Other congenital malformations:** sonography is the initial imaging for **suspected DSD** — müllerian structures and their relation to the urinary tract.
- **Trauma:** abdominal sonography compares favourably with contrast CT and suits **longitudinal follow-up** of renal injury (perinephric haematoma, urinoma); ionising studies with IV or intraluminal iodinated contrast evaluate genitourinary trauma; evaluation is **incomplete without delayed CT views** of parenchyma and collecting system, or **lateral projections** of bladder, bladder neck and urethra.
- **Acute scrotum:** history, examination and scrotal sonography discriminate inflammation, neoplasm, trauma and torsion; nuclear scintigraphy adds little over sonography.
- **Undescended testis:** sonography unreliable — **do not use**, except to seek an inguinal testis in an overweight boy.

COMPARISON — "BOTTOM-UP" VS "TOP-DOWN" IMAGING AFTER FEBRILE UTI

Feature	Bottom-up	Top-down
First test(s)	Ultrasound → VCUG	DMSA (replaces screening ultrasound)
Target	All VUR	APN and/or renal scarring (kidneys at risk)
VCUG	Routine	Only if DMSA abnormal
Weakness	Overtreatment of low-grade VUR	VUR sought only in high-risk (DMSA-abnormal) children

COMPARISON — MODALITY BY CLINICAL QUESTION

Question	First line	Next / problem-solver	Avoid / no role
Hydronephrosis — obstruction?	Sonography	MAG3 diuretic renography (gold standard) ; MRU alternative	CT, IVP
Hydroureteronephrosis ± thick bladder	Sonography	VCUG (PUV, VUR)	—
VUR detection/follow-up	VCUG	CeVUS; direct radionuclide cystography; PIC if occult	Indirect cystography (insensitive)
APN / scarring	—	DMSA (early = APN; ≥6 months = scar)	Grey-scale ultrasound (insensitive); MRI (sedation)
Renal stones	Sonography	CT (burden) under ALARA; KUB	Reflex CT
Renal mass	Sonography	Contrast CT ; MRI (local staging, contralateral lesions)	—
Bladder mass	Sonography (full bladder)	MRI (rhabdomyosarcoma)	—
Abdominal trauma	CT (with delayed phase); sonography	Retrograde urethrogram; CT/conventional cystography	Catheter before urethrogram if meatal blood
Acute scrotum	Clinical ± colour Doppler/whirlpool	Scintigraphy if equivocal and no delay	MRI, CT; any imaging that delays exploration
Undescended testis	Examination	Sonography only in obese child	Routine ultrasound; MRI marginal
DSD	Sonography + genitography	MRI for very complex cases	CT

COMPARISON — NAMED PAEDIATRIC IMAGING SIGNS

Sign	Modality	Appearance	Meaning
Keyhole	Ultrasound (pre/postnatal)	Thick bladder + dilated posterior urethra	PUV / bladder outlet obstruction
Twinkling	Colour Doppler	Colour artefact behind echogenic focus	Calculus (renal, ureteric, bladder)
Whirlpool	High-resolution ultrasound	Twisted spermatic cord	Testicular torsion
Fountain	Colour Doppler	Hyperaemic thickened scrotal wall	Acute idiopathic scrotal oedema
Onion rings	Ultrasound	Concentric hyper/hypoechoic rings, no flow	Epidermoid cyst
Pie-in-the-sky	Cystography	Extreme elevation of bladder neck	Pelvic fracture with extraperitoneal bladder injury

Concept — Each clinical question has **one anatomical and one functional answer**; the paediatric algorithm starts with sonography and escalates only to the functional test that the anatomical finding demands.